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Lab 2 Report

I. Table Comparing Resource Utilization

Resources	Lab 1	Lab 2
Slice LUTs as Logic	8	8
Slice LUTs as Memory	0	8
Slice Registers as Flip Flop	0	0
Slice Registers as Latch	0	0
Number of Bonded IOBs	51	51
Number of Clocks	0	0

II. Comparing Schematics of Lab 1 and Lab 2

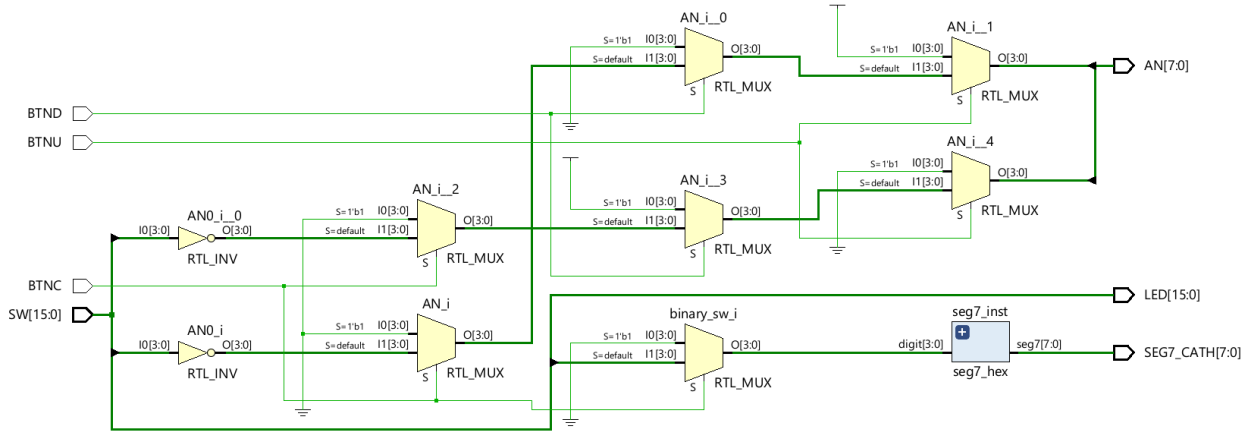


Figure 1: RTL Schematic of lab1_top.

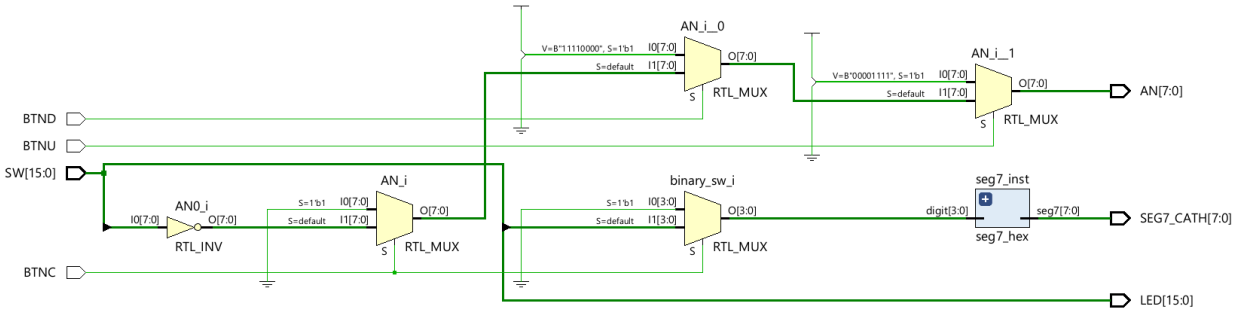


Figure 2: RTL Schematic of lab2_top.

III. Comparing Resource Utilization of Lab 1 and Lab 2

When comparing the results of the synthesis reports, the utilization of resources is the same for Lab 1 as it is for Lab 2. This makes sense from a logical standpoint; the two circuits behaviorally are the same. Neither utilization used any components with memory, including clocks, flip-flops, or latches. This is intentional; the circuits consist only of combinational logic. Instead, the two circuits utilize 8 look-up tables (LUTs) as combinational logic.

From a behavioral and resource utilization standpoint, the circuits in Lab 1 and Lab 2 are identical. However, the schematics slightly differ, with lab1_top having an additional four logic blocks than lab2_top. I believe this is because of the way I programmed lab1_top. When I wrote the VHDL code for the first lab, I separated the assignment of signal AN into two conditional statements: one for the four least significant bits and one for the four most significant bits. However, for the second lab, I used one if/else/then conditional within the process. If I had better optimized my code in Lab 1, I believe the two RTL schematics would be identical.